

Microsoft

Gain the Most Out Of Xamarin.Forms and Visual Studio 2017

David Ortinau

Senior Program Manager of Mobile Developer Tools

Let’s dive into the basics

XAML

Native UI
XAML Compilation

Customizing UI

Routing Effects
Custom Renderers
Deep-Linking
Animations

UI Performance

Startup Time
Layout Optimizations
List Performance
Fast Renderers

XAML

XAML Defined

XAML—the eXtensible Application Markup Language—allows developers to define user interfaces in Xamarin.Forms applications using markup rather than code.

```
<?xml version="1.0" encoding="utf-8"?>
<ContentPage
        xmlns="http://xamarin.com/schemas/2014/forms"
        xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"

        xmlns:local="clr-namespace:Mira.Forms"
        x:Class="Mira.Forms.Mira_FormsPage">

        <Label  Text="Welcome to Xamarin Forms!"
                VerticalOptions="Center"
                HorizontalOptions="Center" />

</ContentPage>
```

```
public class Mira_FormsPageCodeOnly : ContentPage
    {
        public MyPage()
        {
            Content = new Label {
                Text = "Welcome to Xamarin Forms!",
                HorizontalOptions = LayoutOptions.Center,
                VerticalOptions = LayoutOptions.Center
            };

        }
    }
```

Native UI

Forms elements are realized via platform renderers.

	ContentPage	Label
Android	Activity	TextView
iOS	UIViewController	UILabel
UWP	Page	TextBlock

XAML Compiled
Enable XAMLC to Precompile

- Compile time validation
- Better Runtime Performance

```
using Xamarin.Forms.Xaml;
...
[assembly: XamlCompilation (XamlCompilationOptions.Compile)]
namespace PhotoApp
{
        ...
}
```

Customizing UI

Progressive Enhancement

From standard experience to total control

Forms Element > Routing Effect > Custom Renderer

Demo

Deep Links

Launch your app from a search result or URL directly to a targeted Page

- Android
  - Add NuGet Xamarin.Forms.AppLinks
  - MainActivity > AndroidAppLinks.Init(this)
- App
  - Register AppLinkEntry targets
  - Override `OnAppLinkRequestReceived`

Demo

Animation

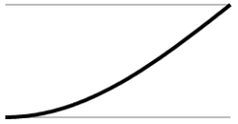
Create meaningful interface feedback

- View Extensions
  - Fade, Layout, Rotate, Scale, Translate
  - Helpers that implement the Animation class
- Animation
  - Build custom animations beyond what you get with View Extensions
- Easing
  - Bounce, Cubic, Linear, Sin, Spring, Custom

Easing Visualized

css+js

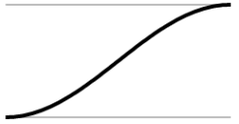
easeInSine



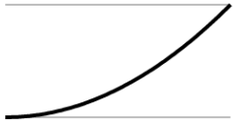
easeOutSine



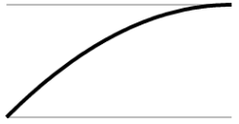
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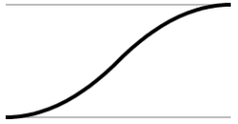
easeInQuad



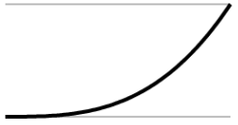
easeOutQuad



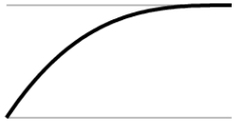
easeInOutQuad



easeInCubic



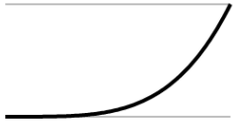
easeOutCubic



easeInOutCubic



easeInQuart



easeOutQuart



easeInOutQuart



easeInQuint



easeOutQuint



easeInOutQuint



easeInExpo



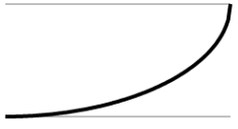
easeOutExpo



easeInOutExpo



easeInCirc



easeOutCirc



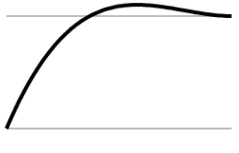
easeInOutCirc



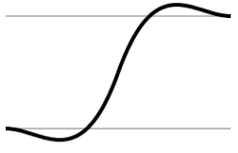
easeInBack



easeOutBack

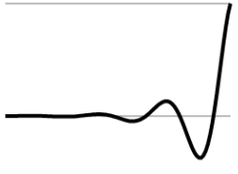


easeInOutBack

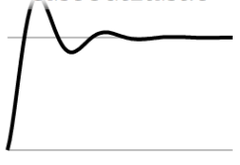


js

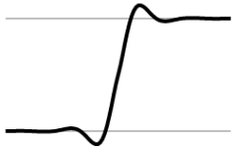
easeInElastic



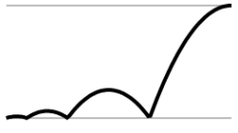
easeOutElastic



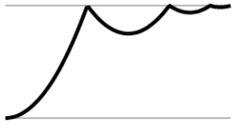
easeInOutElastic



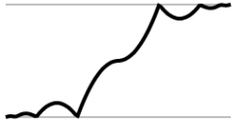
easeInBounce



easeOutBounce



easeInOutBounce



Demo

UI Performance

Startup Time
From launch to first screen

- Do
  - Simplify your first UI
  - Lazy load everything you don’t need immediately
- Don’t
  - Block the UI thread
  - Fetch data from the web
  - Inflate unnecessary UI
  - Preload custom fonts

Demo

Layout Performance
From page to page

- Layout Cycle
  - Begins at top of the visual tree and iterates down to every child
  - Measure and Layout
  - Constrained elements / layouts require fewer cycles
- Invalidation
  - Triggers a new Layout Cycle
  - Element within a constrained layout only triggers a cycle within that parent
- Caching
  - Override InvalidateLayout and OnChildMeasureInvalidated to strictly control triggers

Demo

ListView Performance
Quick Tips

- Fast Scrolling
- Caching Mode
- Constrain and Flatten CellViews

Demo

Thank you.
Questions?

David Ortinau
Senior Program Manager, Xamarin.Forms

@davidortinau



Microsoft

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